Christy Biji

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Education

Whiting School of Engineering, Johns Hopkins University *Aug 2021 - Jan 2023*
MS in Chemical and Biomolecular Engineering

- **Thesis:** Development of a human iPSC derived microglia assay for evaluation of target dependent activation and pharmacological inhibition (Link)

Department of Biotechnology, Savitribhai Phule Pune University *Oct 2020*
B.Tech Biotechnology (Sinhgad College of Engineering)

- **Thesis:** Extraction & Application of Hydrophobin (Link)

Research Experience

The Broad Institute of MIT & Harvard *Cambridge, MA*
Research Associate II, Center for the Development of Therapeutics (CDoT) *April 2023 – Present*

Current Projects

- Generating a morphological dataset across 66 pediatric cancer cell lines using Cell Painting — seeding cells, running liquid handling on the Bravo at 24, 48, and 72-hour timepoints, and setting up high-content imaging independently on the Opera Phenix to feed a deep learning pipeline searching for drugs that push cancer cells back toward a healthy state.
Collaboration: Dr.Gregory Way, University of Colorado.
- Screening a GPCR-focused medicinal chemistry library of 1,200+ compounds using bioluminescence assays; running confirmation assays on primary hits and flagging IC50 outliers to the medicinal chemistry team to drive structure-activity relationship (SAR) decisions.
Collaboration: Stanley Center, The Broad Institute

Completed Projects

- Established HiBit-tagged RH4 and HEK293T cell lines via lentiviral transduction — working independently from existing frozen stocks — to build the experimental foundation for a chromatin displacement assay targeting the PAX3-FOXO1 fusion oncogene in rhabdomyosarcoma; tracked assay performance through luminescence readouts.
Collaboration: Dr. Angela Koehler, MIT.
- Ran high-throughput assays and analyzed dose-response data in Genedata Screener as part of a drug repurposing screen for TFE3 fusion-driven renal cell carcinoma, contributing to the identification of CDK9 inhibition as a candidate therapeutic strategy.
Collaboration: Dr.Viswanathan, DFCI.
- Ran GSK3β biochemical assays on approximately 64 compounds per week, generating IC50 profiles to support an active SAR campaign in bipolar disorder; flagged outlier compounds to the chemistry team when the data suggested something worth looking at more closely.
Collaboration: Stanley Center, The Broad Institute

Merck *Boston, MA*
Master Thesis Researcher, Discovery Neuroscience *June 2022 – Jan 2023*

- Established NLRP3 inflammasome activation protocols using LPS priming followed by ATP and Nigericin stimulation across iMGLs, THP-1, and PBMCs to evaluate CNS-relevant versus peripheral cell line responses in neurodegenerative disease modeling.
- Determined MCC950 IC50 values across cell types — iMGLs (ATP: 98.57 nM, Nigericin: 219.1

nM), THP-1 (16.18 nM), and PBMCs (85.7–204.9 nM at 0% serum) — demonstrating that iMGLs produce pharmacologically valid NLRP3 inhibition profiles comparable to peripheral cell lines.

- Identified a serum-dependent potency shift in PBMCs (IC50 3× higher at 5% versus 0% serum), attributed to MCC950 protein binding, and used this finding to optimize assay conditions for iMGL experiments.
- Confirmed functional inflammasome assembly in iMGLs by visualizing ASC speck formation under ATP and Nigericin stimulation via immunocytochemistry (anti-ASC/DAPI).

Bayview Medical Centre at Johns Hopkins *Baltimore, MD*
Graduate Biospecimen Manager, Part-time *Nov 2021 – May 2022*

- Managed a clinical biorepository for neurological research using Freezerpro LIMS to maintain integrity of CSF, plasma, and serum samples.

Publications

Cancer target discovery enabled by transcriptome-based virtual CRISPR screening *(on bioRxiv)* *[Under Review @ Science]*
Sadagopan A, Li B, Li J, Cui Y, Deodhar R, Yang D, Wu Y, Konda P, **Biji C**, Thapa DR, Thakur M, Weiss CN, Choueiri TK, Cheah JH, Doench JG, Drapkin BJ, Viswanathan SR. bioRxiv [Preprint]. 2025

GPR34 regulation of disease-associated microglial states and responses to physiological stimuli *(on bioRxiv)*
Geller A, Kwon MJ, Simmons SK, Xu Q, Martenis WE, Doman J, Natarajan S, Stalnaker KJ, Zhang J, Adiconis X, Nguyen T, Demers M, Batzli D, Valle-Tojeiro A, **Biji C**, Aryal S, Pribiag H, Depp C, Morshed N, Misri D, Bohannon D, Longhi E, Keshishian H, Carr SA, McKinney D, Gould AE, Lebois E, Weiwer M, Johnson M, Levin JZ, Zhang YL, Sheng M, Kunwar PS. bioRxiv [Preprint]. 2025

Poster Presentation

Biji C, Tomkinson J, Curd J, Skepner A, Way G and Cheah J. ***Generation of a Cell Morphological Landscape in Pediatric Cancer Cell Lines.*** Broad Retreat, December 2025, Cambridge, MA.

Geller A, McKinney D., Longhi E, **Biji, C.**, et al. ***Developing brain penetrant activators of microglia specific GPR34 for modulating function and treating neurodegenerative and psychiatric diseases*** Stanley Center Symposium, September 2025, Cambridge, MA.

Weïwer M., Kyei-Baffour K., Wagner F, Shekar M, **Biji, C.**, et al. ***Inhibition of GSK3a,β rescues motoric and cognitive phenotypes in a preclinical mouse model of CTNNB1 Syndrome.*** Broad Retreat, December 2024, Cambridge, MA.

Computational & Quantitative Skills

- **Machine Learning & Statistical Modeling:** Classification (Logistic Regression, Random Forests), clustering (k-means, Hierarchical), dimensionality reduction (PCA, UMAP), statistical inference, and hypothesis testing, Python (Pandas, Scikit-learn, Matplotlib), R (Tidyverse), MATLAB, Java, Git version control, and Terra cloud computing.
- **Bioimage & High-Content Data Analysis:** End-to-end experience in high-content screening workflows, including quantitative morphological profiling (CellProfiler, ImageJ, Cellpose) and large-scale data analysis (Genedata Screener, LIMS).

- **Experimental Biology Foundation:** Mammalian cell culture, Cell Painting, ELISA, High Throughput multiplexed screening in 384-well format, Dose-response profiling, immunocytochemistry (ICC), BSL2+/BSL2 and with automated screening instrumentation (Opera Phenix High Content Screening System, Beckman Coulter Echo, Biotek, Pherastar).

Projects

BoomExtractor - AI-Powered Scientific Workflow Visualization *BoomExtractor*
(Broad Hackathon 2025 (Slides))

- **Awarded BroadHacks Best Overall Project and the People's Choice award.** (Link)
- Led the end-to-end development of an AI-driven tool to accelerate research comprehension by automatically generating scientifically-validated experimental workflow diagrams from papers using **Python, Gemini AI,** and **React**.

MIT IDSS — Data Science and Machine Learning *Jan 2025*
(Course projects from the MIT IDSS program)

MIT-IDSS-Recommendation-System *Project Link*

- Engineered an end-to-end recommendation system by benchmarking collaborative filtering models (User-User, SVD) and optimizing the top-performing **SVD model** with `GridSearchCV` to achieve a final **RMSE of 0.88**.

MIT-IDSS-Lead-Scoring-Model *Project Link*

- Developed an end-to-end lead scoring pipeline by tuning multiple classifiers (**Random Forest, Decision Tree**) with GridSearchCV to optimize for recall (82%), enabling data-driven prioritization of high-potential sales leads.

MIT IDSS Project: Customer Personality Segmentation *Project Link*

- Applied unsupervised learning (**K-Means**, PCA) to segment a retail customer database, identifying and profiling distinct high-value and low-value personas to inform targeted marketing and retention strategies.

Engineering Lungs with mtDNA-Edited iPSCs *Technical Report*
(Tissue Engineering Project)

- Proposed a novel framework for lung bioengineering combining decellularized scaffolds with autologous, **mtDNA-edited iPSCs** to prevent transplant rejection and expand the donor organ pool.

Knockout NEU3 gene target with CRISPR-Cas9 *Project Report*
(Cell & Tissue Engineering Lab)

- Engineered a CRISPR-Cas9 gene-editing tool to knock out the sialidase Neu3 in CHO cells, a strategy aimed at increasing the sialylation and therapeutic half-life of recombinant hIFNγ.

Leadership and Services

Treasurer, Women at Broad Jan 2026
The Broad Institute

Spot Award - awarded for excellence in the service of the department May 2025
CDoT, The Broad Institute of MIT and Harvard

Graduate Student Liaison Committee May 2021
Whiting School of Engineering - Johns Hopkins University

Courses and Certification

Data Science and Machine Learning: Making Data-Driven Decisions Jan 2025

MIT Institute for Data, Systems, and Society (IDSS) (Link)

Mathematical Statistics (Graduate-Level Coursework) Dec 2024

Harvard University Extension School Grade: A (Transcript Link)

Introduction to the Biology of Cancer July 2020

Johns Hopkins University, Coursera (Link)

Genes and the Human Condition (From Behavior to Biotechnology) June 2020

University of Maryland, Coursera (Link)

Antimicrobial Resistance - Theory and Methods June 2020

Technical University of Denmark, Coursera (Link)